INCH-POUND

MIL-I-83026D

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SUPERSEDING
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27 MAY 1977

MILITARY SPECIFICATION

INDICATORS, RATE OF FLOW, FUEL

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. Scope

1.1 Scope. This specification covers the requirements for certain types of hermetically sealed integrally lighted indicators for indicating the flow rate of engine fuel in pounds per hour (pph). The indicators are designed to be compatible with a particular transmitter which provides the proper signal format and range for the specific indicator.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

Basic Type	Repeater Type	Integral Lighting	Flow Range (PPH)
EFU-26/A-7	EFU-43/A-7	Red	0-15,000
EFU-26/A-8	EFU-43/A-8	White	0-15,000
EFU-26/A-13	EFU-43/A-13	White	0-15,000

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145-5990 by using the self-addressed Standardizaton Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-E-5400	Electronic Equipment, Airborne, General Specificaton For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Spcification For
MIL-C-14806	Coating, Reflection Reducing, For Instrument Cover Glasses And Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specificaton For
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic
MIL-T-83027	Transmitter, Rate Of Flow, Fuel Tru-98/A

STANDARDS

Federal

FED-STD-595	Color

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-701	Lists Of Standard Semiconductor Devices
MIL-STD-704	Electric Power, Aircraft, Characteristics And Utilizations Of

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MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-882	System Safety Program For Systems And Associated Substsystems And Equipment: Requirements For
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards And Specifications, Order Of Preference For The Selection Of
MIL-STD-2073-1	DoD Material Procedures For Development And Application Of Packaging Requirements
MS3113	Connectors, Receptacle, Electric, Solder Mounting, Miniature Bayonet Coupling
MS3126	Connector, Plug, Electric, Crimp-Type, Straight, Bayonet Coupling
MS33558	Numerals And Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

HANDBOOKS

AFSC Design Handbook

DH 1-6	System Safety

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

(Copies of MIL-STD-810C is available from Director, Naval Publicatons and Printing Service Office, 700 Robbins Ave Bldg 4 Section D, Philadelphia PA 19111.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be address to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this detail and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the test of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified product list at the time of award of contract (see 4.4 and 6.5).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Preliminary list of electronic parts. A preliminary list of electronic parts shall be furnished directly to the contracting activity in accordance with MIL-E-5400 prior to ordering or fabricating parts for the production quantity. The parts shall conform to MIL-E-5400. Nonstandard parts shall be approved by the contracting activity prior to the fabrication of the completed item. Whenever practicable, nonstandard electronic parts shall not constitute more than 30 percent of the total quantity of electronic parts used in the construction of the indicator.

3.4 Materials

3.4.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.4.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator, except where ferrous materials are essential.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.4.3.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Toxic and Corrosive fumes. The indicator shall be of such material and construction that when the material is burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.4.5 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.6 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

3.5 Design and construction. Unless otherwise specified, the requirements are applicable to all types of indicators covered by this specification.

3.5.1 Function. The indicator shall operate from a second harmonic input that shall be received from its respective transmitter. The EFU-26/A series indicators shall include a constant frequency power supply to drive the respective transmitter.

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The EFU-26/A series indicators are repeater types only for use in dual place aircraft or when another fuel flow display is required.

3.5.1.1 Second harmonic input. The EFU-26/A-7, /A-8, and/A-13 and the EFU-43/A-7, /A-8, and /A-13 indicators shall include all of the circuitry and mechanism necessary to convert a second harmonic signal from a TRU-98/A transmitter conforming to MIL-T-83027 to a pointer indication of fuel flow rate in pounds-per-hour of aviation fuel. The indicator shall be designed to operate on the second harmonic system principle in which the moving element consists of nonenergized permanent 2-pole magnets. The toroidally wound stator is energized with 115V 400-Hz single-phase a-c power. Remote indicaton of the angular position of the turbine in the TRU-98/A transmitter is obtained through the use of the coil and core operating on the permanent magnet rotor (turbine). The direct-current resistent of the pickoff coil sections shall be between 290 and 350 ohms per section when the coil temperature is 25° ±5°C.

3.5.2 Input signals

3.5.2.1 Transmitter rotation. The TRU-98/A transmitter is calibrated so that the turbine will rotate 307° for a full-scale reading. For an increasing flow rate, the turbine shall rotate in a clockwise direction when viewed from the inlet end.

3.5.2.2 Power. The indicators shall be designed to operate from 115V ±10 percent 400 Hz ±10 percent single-phase power as specified in MIL-STD-704 for equipment having Category B characteristics. In addition, the EFU-26/A-13 and EFU-43/A-13 indicators shall operate from power having a frequency variation as great as ±20 percent except that the ratio of frequency to voltage shall deviate from normal (400/115) by more than ±22 percent.

3.5.2.2.1 Power consumption. Indicator power consumption (including the power supply) shall not exceed 50 va for the EFU-26/A series indicators with the transmitter load connected but excluding the lighting circuit and 15 va for EFU-43/A series indicators.

3.5.3 Output signals

3.5.3.1 Power supply. The EFU-26/A series indicators shall supply either 34 ±4V rms 8-Hz ±0.05 percent 2-phase power or 65 to 126V 400 ± Hz single-phase a-c power, but not both, to the TRU-98/A transmitter conforming to MIL-I-83027. The power supply shall meet the applicable tolerances of table I. The output of the power supply shall be either untuned or broadly tuned and shall operate one transmitter.

3.5.3.2 Power supply warmup. After a warmup period of not more than 1 minute, the power supply shall meet the tolerances specified in table I over an operating range of -54° to 71°C.

3.5.4 Case. The case shall conform to MS33639 for the 2-inch size, shall be finished in durable dull black, and shall be hermetically sealed. The seal shall be so designed and constructed that it will not be dependent upon materials that might be affected by the action of any atmospheric conditions to which the case may be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than 0.031 inch. The case length, dimension L, shall not exceed 7.312 inches for the EFU-26/A series indicators and 3.5 inches for the EFU-43/A series indicators, exclusive of the electrical receptacle. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools unless they are

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TABLE I. Indicator tolerances.

PPH TOLERANCES				
TEST	0-15,000 Range		0-15,000 Range EFU-26/A-13 and EFU-43/A-13	
	Low 0-6,000	High Above 6,000	Low 0-4,000	High Above 4,000
Position error	30	85	20	100
Hysteresis	30	85	20	100
Vibration				
Pointer oscillation (Double amplitude)	45	125	30	150
Pointer variation	30	85	20	100
Friction error				
Room and high temperature	30	85	20	100
Low temperature	45	125	30	150
Voltage and frequency variation	40	110	27	125

TOLERANCE (PERCENT OF RATED FREQUENCY)		
Power supply	8-Hertz power	400-Hertz power
Room temperature	±0.05	±0.25
Voltage & frequency	±0.07	±0.33
High temperature	±0.09	±0.50
Low temperature	±0.09	±0.50
Vibration resonance	±0.09	±0.50
Room temperature following reliability	±0.10	±1.00

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approved by the contracting activity.

3.5.4.1. Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium shall be 12 to 14.7 psi. Other concentrations of helium shall not be used, except with prior approval of the contracting activity.

3.5.4.2 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface.

3.5.5 Dials. The dials of the indicators shall conform to figure 1. The numerals and letters shall be in accordance with MS33558. The diameters of the dial presentations shall be 1.75 inches, minimum.

3.5.6 Marking. The marking dimensions of the indicator dials shall be in accordance with table II.

Table II. Dial Marking dimensions.

Marking (15,000 pph scale)	Height or Length (Inch ±0.156)	Width of Line or Graduation (Inch ±.005)
Scale numerals 0 through full scale	0.156	0.031
1,000 pph graduations	0.156	0.031
500 pph graduations	0.125	0.020
100 pph graduations	0.094	0.010
Lettering PPH X 1000	0.062	0.010
Lettering FUEL FLOW	0.094	0.016

3.5.7 Lighting. The indicator dial and pointer shall be either integrally red lighted in accordance with MIL-L-25467 or integrally white lighted in accordance with MIL-L-27160, as applicable.

3.5.8 Pointer. The pointer shall be in accordance with MS33585-8, except that it shall be suitable for integral lighting as shown on figure 1.

3.5.9 Cover glass. The cover glass shall be clear, flat, and free from flaws which would interfere with the normal reading of the instrument. The glass shall not be fluorescent when exposed to ultraviolet light or to rays in the 363 millimicron region. All reflecting surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-14806.

3.5.9.1 Dial to cover glass distance. The distance between the inner surface of the cover glass and the indicator dial shall be the minimum practicable and shall not exceed 0.188 inch. The ends of the major graduations shall be visible when viewed from any direction within a 30°angle from a line perpendicular to the center of the indicator dial.

3.5.10 Compatibility. All indicators with identical type designations shall be interchangeable. The EFU-26/A-7, /A-8, EFU-43/A-7. and /A-8 indicators shall be completely compatible with previously manufactured TRU-98/A-4 transmitters. The

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EFU-26/A-13 and EFU-43/A-13 indicators shall be completely compatible with the TRU-98/A-7 transmitter.

3.5.11 Damping. The indicator shall be capable of attenuating small oscillatory transmitter output signals. The indicator damping shall minimize indicator response to fluctuations in fuel flow of ±1.5 percent of full scale at frequencies at 2.0 Hz and higher.

3.5.12 Electrical receptacle. Each indicator shall be provided with an MS3113H12-10P hermetically sealed receptacle conforming to MIL-C-26482.

3.5.12.1 Wiring. All indicators shall be wired in accordance with figure 2.

3.5.13 Transistors and diodes. Transistors and diodes used in the construction of the indicator shall be of the silicon type designed for continuous operation in a 70°C ambient temperatue in accordance with MIL-STD-701. Circuit design shall be such that no electronic component will be operated outside its rated characteristics with 115V 400 Hz power input in accordance with 3.5.2.2 and with indicator input signals as specified herein. No electronic device shall break down while the system (indicator and transmitter) or indicator alone is properly connected and subjected to the power variations in accordance with 3.5.2.2.

3.5.14 Maintainability. The indicator shall be so designed as to provide ease of maintenance which shall include the following:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Rapid and positive recognition of equipment malfunction or marginal performance.

c. Rapid and positive identification of the replaceable defective part, assembly, or component.

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

e. Optimum accessibility in all components requiring maintenance inspections, removals, or replacements.

3.5.15 Reliability. The indicator shall have a lower test mean-time-between-failure (MTBF) ( ) of 1200 hours with an upper test MTBE ( ) of 1800 hours, as defined in MIL-STD-781C, demonstrated to a 90 percent confidence level."

3.5.16 Longevity. The indicator shall have an operating life span of not less than 2,500 hours before wearout failures occur or the indicator demonstrates a decrease in MTBF. Parts requiring replacement within this period of time and the

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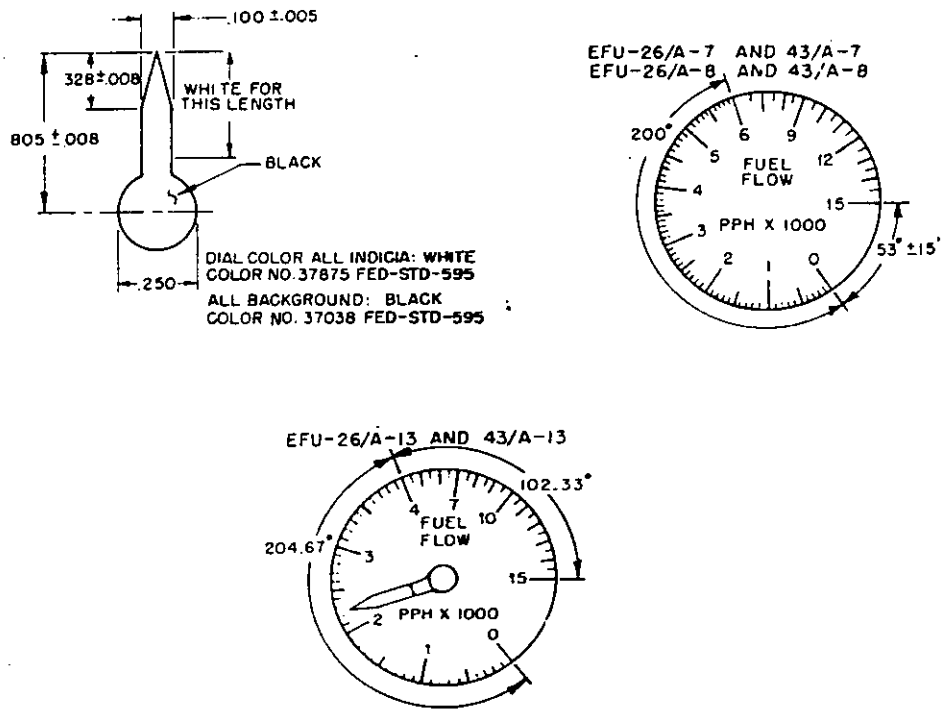


Figure 1. EFU-26/A and 43/A fuel flow indicator scale and pointer presentations.

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normal operating period of such parts shall be reported by the contractor to the contracting activity.

3.5.17 Synthetic jewel bearings. When sapphire or ruby synthetic jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.5.18 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.5.19 Range. The indicator presentations shall be split range as shown on figure 1. The scales shall be linearly graduated in the contracted and expanded ranges within 15 minutes of arc between adjacent scale graduations.

3.5.20 System Safety. The indicator shall be so designed that the design will not result in a Category III or Category IV hazard as defined in MIL-STD-882, paragraph entitled Hazard Level, for the predicated operating life specified in 3.5.16.

3.6 Performance. The indicators shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperatures - operating temperatures ranging from -54° to 71°C and storage temperatures ranging from -62° to 71°C.

b. Vibration - vibration from 5 to 500 Hz at double amplitude of .00016 to 0.250 inch for 9 hours.

c. Thermal shock - eight cycles of alternate immersions in water at 85°C and 5° for 30 minutes.

d. Salt fog - salt fog for a period of 48 hours.

e. Voltage and frequency variation - voltage and frequency variation ranging from 104V 360 Hz (320 Hz for the EFU-26/A-13 and EFU-43/A-13 indicators) to 126V 440Hz (480 Hz for the EFU-26/A-13 and EFU-43/A-13 indicators) as specified in 4.6.12.

f. Altitude - absolute pressure equivalent to 100,000 feet.

g. Shock - up to 15g.

h. Frequency cycling from 320 to 480 Hz.

i. Voltage surges to 195V.

3.6.1 Fogging. There shall be no fogging of the face of the indicator glass when the indicator is energized and kept in a 70° ±2°C controlled ambient temperature for a minimum of 1 hour and the glass face then rubbed with an ice cube for a period of 1 to 2 minutes.

3.6.2 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 as specified for class A1 equipment.

3.6.3 Magnetic effect. The indicators, operating or not operating, shall not create a magnetic field with an intensity that will deflect a short bar magnetic compass more than 3°.

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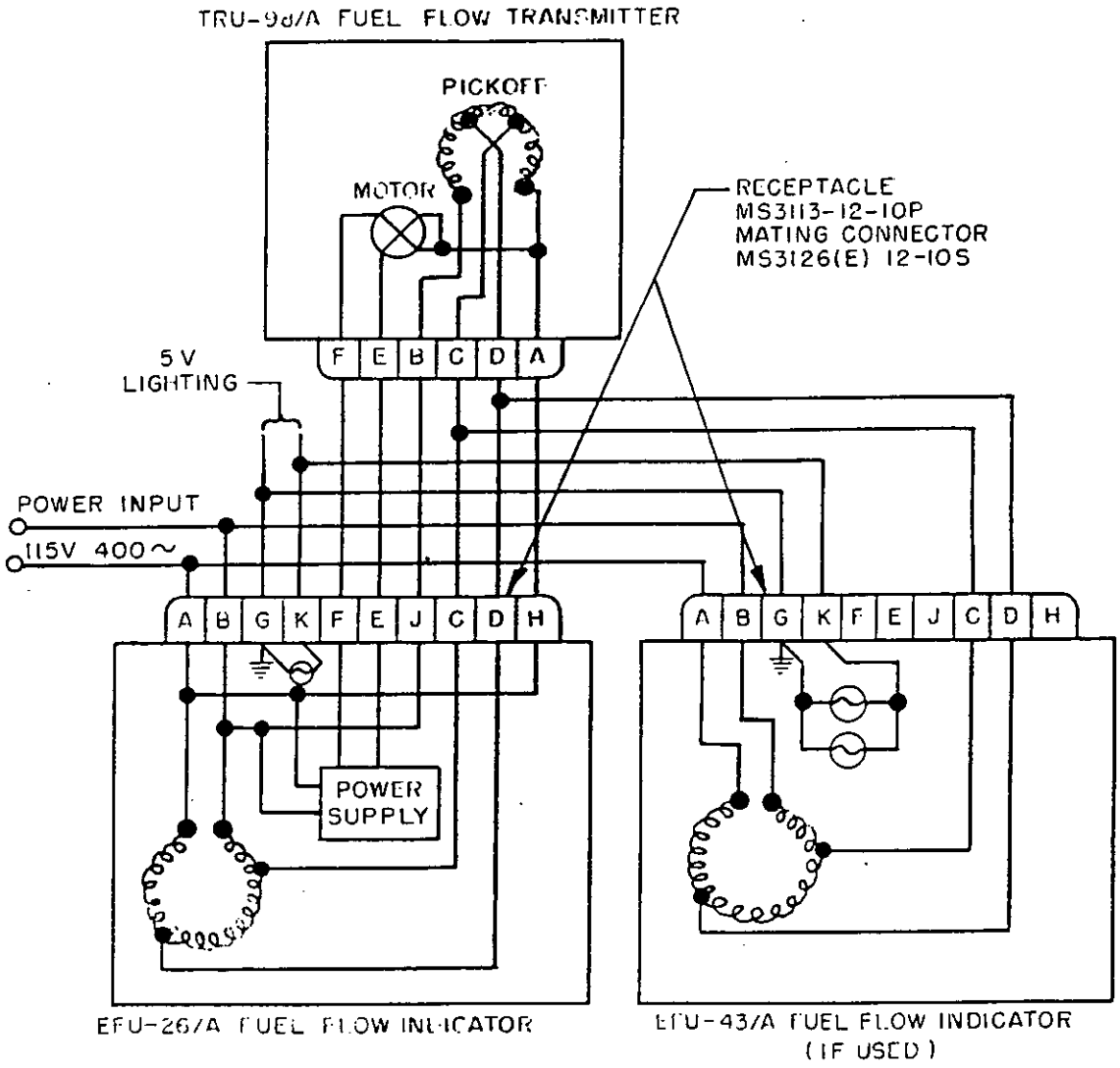


Figure 2. System connection diagram for EFU-26/A and 43/A indicators.

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3.6.4 Response. The indicator shall show a rate indicaton from zero to half scale or from half scale to zero in less than 3 seconds at room temperature.

3.6.5 Accuracy. Scale error, hystersis, and friction error shall be within the tolerances specified in tables I and III.

3.6.6 Dielectric strength. The indicator shall withstand applications of 500V at a commercial frequency applied between all pins not used for grounding the case for a period of 5 seconds before the indicator is sealed and 200V after the indicator is sealed.

3.6.7 Damping. The pointer shall oscillate with a total excursion less than 2 percent when the input signal to the indicator is oscillated about the test points nearest to the 5, 10, 50, and 90 percent of full scale test points with a 3 percent of full scale double amplitude at a frequency between 2.0 and 2.4 Hz.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Weight. The weight of the EFU-26/A series indicators shall not exceed 1.35 pounds, and the weight of the EFU-43/A indicators shall not exceed 1.0 pound.

3.9 Indentification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. Metal decals shall not be used.

3.9.1 Nameplate. A nameplate shall be provided on the rear of the case. When applicable, the information shall conform to MIL-STD-130. Metal decals shall not be used.

3.9.2 Electrical schematic. An electrical schematic showing the function of each connector pin shall be permanently attached to the indicator. An approved metal decal may be used provided it is permanently attached and cannot be peeled back with the fingernail.

3.10 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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TABLE III. Fuel Flow Rate Tolerance in ±PPH (EFU-26/A and EFU-43/A Indicators.

Fuel Flow Test Points PPH	0-15,000 Range		0-15,000 Range, /A-13	
	Room Temp & High Temp	Low Temp	Room Temp & High Temp	Low Temp
200	50	75	33	49
500	48	72	31	45
1,000	45	67	30	42
1,500	44	66	27	41
2,000	42	63	27	41
2,500	42	63	27	41
3,000	42	63	27	41
3,500			27	41
4,000	42	63	27	41
5,000	45	67	160	240
6,000	45	67	160	240
8,000	130	195	160	240
10,000	130	195	160	240
12,000	130	195	160	240
14,000	160	240	160	240
15,000	170	255	160	240

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Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4)

b. Quality conformance inspection (4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections be performed in accordance with the test conditions specified in 4.3.1 in this specification.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3.2 Vibration. Unless otherwise specified, the indicator shall be gently vibrated before a test reading is taken. Vibration shall be applied by electromechanical vibrators at a level and frequency approved by the contracting activity.

4.3.3 Vibration stand. Whenever a vibration stand is specified, the stand shall vibrate at all frequencies between 5 and 500 Hz.

4.3.4 Test position. Unless otherwise specified, the indicator shall be tested with the case in a horizontal position with the scale in the normal viewing position.

4.3.5 Master transmitter. The indicator shall be tested with a master transmitter that generates a signal in accordance with 3.5.2.1. Sufficient tests shall be performed to determine that the transmitter signal is not in error in excess of 0.10°.

4.3.6 Tolerance. Test tolerances shall be as specified in tables I and III.

4.3.7 Test power. Unless otherwise specified, the indicator shall be tested with 115 ±3V, 400 ±2 Hz single-phase, a-c power, and 5.0 ±0.1V a-c or d-c power for lighting.

4.3.8 Test load. Unless otherwise specified, the EFU-26/A series indicators shall be tested while connected to a transmitter or simulated transmitter load. The test wiring that may be used is shown on figure 2.

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4.4 Qualification inspection.

4.4.1 Test samples. Test samples shall consist of 10 indicators representative of the production equipment. Three of the samples shall be subjected to all of the tests in paragraphs 4.6.1 through 4.6.29, and seven samples shall be subjected to the tests of 4.5.1, 4.6.30, and 4.6.31. The samples shall be identified with the manufacturer's part number and such other information as required by the contracting activity.

4.4.1.1 Disposition of test samples. Upon completion of testing, all test samples shall be submitted to the contracting activity in the as-is condition. Unless otherwise specified (see 6.2), the qualification inspection samples will be retained by the contracting activity.

4.4.2 Qualification inspections. The qualification inspections shall consist of all the tests specified under 4.6, Unless otherwise specified (see 6.2), the tests shall be conducted in the order listed.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests.....See 4.5.1

b. Sampling plans and tests.....See 4.5.2

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product.....See 4.6.1

b. Dielectric strenght test.....See 4.6.2

c. Scale error test.....See 4.6.3

d. Friction error test.....See 4.6.4

e. Hysteresis test.....See 4.6.5

f. Response test.....See 4.6.6

g. Sealing test.....See 4.6.7

h. Burn-in test.....See 4.6.8

i. Fogging test.....See 4.6.9

j. Lighting (individual tests of MIL-L-25467 or MIL-L-27160, as applicable)

k. Damping test.....See 4.6.10

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. A minimum of three indicators shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests as described under 4.6:

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a. Individual test.....See 4.5.1

b. Vibration error test.....See 4.6.11

c. Voltage and frequency variation test..See 4.6.12

d. Voltage surge test.....See 4.6.14

e. Low temperature test.....See 4.6.15

f. High temperature test.....See 4.6.16

g. High altitude - low temperature test..See 4.6.17

h. Position error test.....See 4.6.18

i. Power consumption test.....See 4.6.19

j. Lighting (sampling plan A tests of MIL-L-25467 or MIL-L 27160, as appliable).

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three indicators shall be selected at random from the first 10 items produced on each contract or order and subjected to the following tests in the order specified (unless otherwise approved by the contracting activity) and as described under 4.6. If any one of the samples specified in sampling plan A is selected for sampling plan B, the sampling plan A tests shall not be repeated on that sample.

a. Sampling plan A tests.....See 4.5.2.1

b. Magnetic effect test.....See 4.6.20

c. Frequency cycling test.....See 4.6.13

d. Low temperature exposure test.....See 4.6.21

e. High temperature exposure test.....See 4.6.22

f. Salt fog test.....See 4.6.23

g. Thermal shock test.....See 4.6.24

h. Shock test.....See 4.6.25

i. Electromagnetic interfenence test.....See 4.6.26

j. Lighting (sampling plan B tests (including life test) of MIL-L-25467 or MIL-L-27160)

k. Vibration endurance test.....See 4.6.27

l. Compatibility test.....See 4.6.29

4.5.2.2.1 Test schedule. When sampling plan B tests are not specifically waived on a contract, the tests shall begin as soon as the first 10 items have been produced but before any production deliveries have been made. The test data shall

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be submitted to the contracting activity no later than 16 weeks after starting the test program.

4.5.2.3 Rejection and retest. When an item selected from a production run fails to meet the specification, indicators still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, tests shall be repeated as determined by the contracting activity.

4,5,2,3,1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of indicators on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.2.4 Sampling plan C. Seven indicators shall be selected at random from the first 14 produced and subjected to the following tests as described under 4.6. Rejection shall be in accordance with 4.6.30.6.

a. Individual tests.....See 4.5.1

b. Reliability test.....See 4.6.30

c. Longevity test.....See 4.6.31

4.5.2.4.1 Reliability test program. Within 90 days after receipt of the contract or purchase order and 30 days prior to the scheduled date of the initiation of testing or at such later date as the detailed reliability test program to the contracting activity for approval. Testing shall begin only after formal approval of this program.

4.5.3 Defects in items already accepted. The investigation of test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, markings, dimensions, and weight.

4.6.2 Dielectric strenght test. Prior to sealing, 500V at a commercial frequency shall be applied between certain pins and the case for a period of 5 seconds. Only those pins not used for grounding the components to the case shall be used. After sealing, 200V at a commercial frequency shall be applied between only hose pins not used for grounding and the case for a period of 5 seconds. There shall be no insulation breakdown. All lighting pins shall be considered grounded for this test.

4.6.3 Scale error test. The indicator shall be connected to a master transmitter. The transmitter shall be set at angles equivalent to each of the test points in table III. The indicator shall read within the tolerances specified therein for the applicable range. The output of the indicator power supply shall also be within the tolerances specified in 3.5.3.1 and tables I.

4.6.4 Friction error test. The indicator shall be tested for friction at the scale

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error test points nearest to 5, 10, 50, and 90 percent of full scale. The master transmitter shall be made to slowly approach the test point in an ascending direction. After the transmitter has been held at the test point for approximately 5 seconds, the indicator pointer reading shall be recorded. The indicator shall then be gently vibrated to establish a stable condition and the reading shall be recorded. The difference between the two readings shall be recorded as friction and shall not exceed the specified tolerance in table I.

4.6.5 Hysteresis test. The master transmitter shall be made to slowly approach the test point nearest to the 20, 50, and 90 percent of full scale and the indicator shall be vibrated. Readings shall be taken at the test point when it has been approached in this manner from the upscale and downscale directions. The difference in these readings shall not exceed the tolerances specified in table 1.

4.6.6 Response test. At 25°C, the indicator shall respond and indicate a half-scale reading within 1 percent when a step input of 153.5 angular degrees is applied from zero angular degrees (zero pph) to 153.5 angular degrees and a full-scale reading within 1 percent from 153.5 angular degrees to 307 degrees of arch in not more than 3 seconds. The indicator shall not be energized for a least 3 minutes prior to this test.

4.6.7 Sealing test. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differenctial of 1 atmosphere.

4.6.8 Burn-in test. The indicator shall be placed in a chamber and the temperature maintained at 50°C for a minimum of 24 hours. Power shall be applied in accordance with 4.3.7 throughout the test while maintaining a constant signal input equivalent to a half-scale indication. At least 8 hours of this shall be performed during normal working hours, and observations shall be made and recorded at least once every 2 hours throughout the 8-hour period. At no time shall the indicator display or the power source voltage output change by more than one-half of the tolerance specified for the test point in the applicable test tolerance table. Pointer oscillation or erratic movement shall be cause for rejection. The indicator shall be returned to room temperature for a minimum of 2 hours after which it shall meet the test specified in 4.6.3.

4.6.9 Fogging test. The indicator shall be energized and kept in a 70° ±2°C controlled ambient temperature for a minimum of 1 hour. At the end of this period and while still at the high temperature, an ice cube shall be rubbed on the face of the indicator glass for a period of 1 to 2 minutes. The glass shall then be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water and oil fog. Any evidence of fogging shall be cause for rejection. This test may be conducted in conjunction with the high temperature scale error and friction tests.

4.6.10 Damping test. The input signal to the indicator shall be made to oscillate about the test points nearest to the 5, 10, 50, and 90 percent of full scale test points with a double amplitude of 3 percent of full scale and at a frequency between 2.0 and 2.4 Hz. The pointer shall oscilllate with a total excursion less than 2 percent of full scale.

4.6.11 Vibration error test. The indicator shall be subjected to a vibration of 0.150 inch double amplitude from 5 to 12.5 Hz and ±1.25g from 12.5 to 500 Hz and return in approximately 15 minutes in each of the three mutually perpendicular axes.

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With a signal input of approximately half scale, the pointer oscillation and variation shall not exceed the tolerances specified in table 1.

4.6.12 Voltage and frequency variation test. This test shall be conducted with the indicator connected to a load impedance as shown on figure 2. Power shall be applied to obtain the following combinations of voltage and frequency: 104 ±1V 360 ±2 Hz (320 ±Hz for the EFU-26/A-13 and EFU-43/A-13 indicators); 126 ±1V 360 ±2 Hz (and 117 ±1V 320 ±2 Hz for the EFU-26/A-13 and EFU-43/A-13 indicators); 126 ±1V 440 ±2 Hz (480 ±2 Hz for the EFU-26/A-13 indicator); and 104 ±1V 440 ±2 Hz and 113 ±1V 480 ±Hz for the EFU-26/A-13 indicator. For each voltage-frequency combination, the accuracy of the indications shall be determined by positioning the master transmitter to obtain simulated flow rates of approximately 5, 10, 50, and 90 percent of full scale. The accuracy of the indicated flow rates and the output of the regulated power supply shall be recorded as specified in table 1. The error of the master transmitter shall be determined at the specified test points and with specified voltages and frequencies prior to testing. The transmitter error shall be algebraically subtracted from the indicated flow rates to determine the indicated error.

4.6.13 Frequency cycling test. With the indicator connected as specified in 4.3.8, the voltage shall be adjusted to 115 ±1V and the frequency cycled from 360 to 440 Hz (320 to 480 Hz for the EFU-26/A-13 andd EFU-43/A-13) in accordance with the following schedule with approximately half-scale flow:

a. Start cycle at 360 Hz (320 Hz for the EFU-26/A-13 and EFU-43/A-13), increase frequency to 440 Hz (480 Hz for the EFU-26/A-13 and EFU-43/A-13) in 38 seconds lineally with time.

b. Return frequency to 360 Hz (320 Hz for the EFU-26/A-13 and EFU-43/A-13) in 2 seconds or less.

c. Increase frequency to 440 Hz (480 Hz for the EFU-26/A-13 and EFU-43-A-13) in 58 seconds lineally with time.

d. Return frequency to 360 Hz (320 Hz for the EFU-26/A-13 and EFU-43/A-13 in 2 seconds.

This 100-second cycle shall be repeated continuously for 10 hours at the rate of 36 cycles per hour. Power shall then be removed for a period of 2 ±0.050 hours and shall be reapplied at 121 ±1V 320 ±2 Hz for 1 hour. Immediately following this 1-hour period, the indicator shall be frequency cycled for 1 hour in accordance with the schedule indicated in a, b, and c except the frequency range shall be increased to between 320 and 480 Hz. The indicator shall then be subjected to the scale error test at room temperature.

4.6.14 Voltage surge test. The indicator shall be connected as shown on figure 2. This test shall be made at approximately 5, 50, and 90 percent of full scale. Rated power shall be applied for 30 minutes. The power input shall then be adjusted to 108 ±1V 480 ±2 Hz for 1 hour and adjusted immediately to 121 ±1V 480 ±2 Hz for 30 minutes. At the end of this 2-hour operating period, the power input shall be adjusted to 115 ±1V 320 ±2 Hz. A voltage surge shall then be applied to provide an rms peak voltage of 160 ±5V and return to 115 ±1V. The duration of the surge,

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including rise and decay time, shall be 0.09 ±0.005 second. The frequency shall then be increased to 480 ±2 Hz and a voltage surge shall be applied to provide a monentary rms voltage of 195 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.07 ±0.005 second. At the end of this test, the voltage and frequency shall be adjusted to 115 ±1V 400 ±2 Hz and a scale error test at room temperature shall be conducted. The errors shall not exceed the room temperature tolerances of tables I and III.

4.6.15 Low temperature test. The indicator, not energized, shall be subjected to a temperature of -54° ±2°C for a period of 4 hours. While at this temperature, the master transmitter shall be set at zero pph. Power shall be applied to the indicator. The output of the power supply shall be monitored for compliance with 3.5.3.1 and table I. At the end of a 10-minute period, the master transmitter shall be set to the half-scale test point, and the time for the indicator to reach this point shall not exceed 25 seconds. The indicator shall then be subjected to the scale error and friction test and shall meet the scale and friction error tolerances specified in tables I and III. At the conclusion of this test and while the indicator is still in a -54°C ambient, the response test shall be conducted. The response time at -54°C shall not exceed 10 seconds. This test shall be conducted immediately after the low temperature scale error test and prior to application of power to the indicator for an accumulation of 40 minutes.

4.6.16 High temperature test. The indicator, while energized, shall be subjected to a temperature of 71° ±2°C for a period of 4 hours and tested at this temperature for scale and friction errors. The indicator shall meet the tolerances of table I and III. At the conclusion of this test and while the indicator is still in a 71°C ambient, the response test shall be conducted. The response time at 71°C shall not exceed 3 seconds. This test shall be conducted immediately after the high temperature scale error test. The output of the power supply shall be checked for compliance with 3.5.3.1 and table I.

4.6.17 High altitude-low temperature test. The indicator shall be subjected to a temperature of -55° ±2°C for a period of 4 hours. During the last hour of this period, the ambient pressure shall be reduced to an altitude of approximately 100,000 feet. While still at this pressure and temperature, the indicator shall be subjected to the scale error test specified in 4.6.3. The scale error shall not exceed the tolerances specified in table III. There shall be no damage to the indicator as a result of this test.

4.6.18 Position error test. The scale error test shall be repeated at the scale error test points nearest to 5, 10, 50, and 90 percent of full scale with the instrument turned 90° right, 90° left, face up, and face down. The change in scale error shall not exceed the tolerances specified in table I.

4.6.19 Power consumption test. The indicator shall be connected with a load impedance as shown on figure 2 and power shall be applied. The power consumption shall not exceed that specified in 3.5.2.2.1, exclusive of lighting power and power consumed by the master transmitter.

4.6.20 Magnetic effect test. The indicator, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5.50 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field

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intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°. The test shall be repeated with the indicator operating at rated voltage.

4.6.21 Low temperature exposure test. The low temperature exposure test shall be conducted in accordance with MIL-STD-810C, Method 502.1 except that after the 48-hour exposure at -62°C, the temperature shall be raised to -54°C and maintained for 24 hours. Upon completion of the 24-hour period at -54°C and while still at this temperature, the indicator shall be subjected to the scale error test and shall meet the tolerances specified in 4.6.15. The EFU-26/A series indicators shall meet the power supply warmup requirements of 3.5.3.2. The indicator shall then be maintained at room temperature for a period of not less than 4 hours and subjected to the scale error at room temperature test. The scale errors shall not exceed the tolerances specified. The indicator shall be examined for deterioration or damage due to extreme temperature exposure.

4.6.22 High temperature exposure test. The high temperature exposure test shall be conducted in accordance with MIL-STD-810C, Method 501.1. After the 48-hour period and while at 71°C, the indicator shall be tested for scale error. During the 48-hour period, the indicator shall be subjected to 71°C ambient for 2 hours while not energized. It shall then be energized and shall meet the warmup perods specified in 3.5.3.2. for the EFU-26/A series indicators. The indicator shall remain energized for the other 46 hours of the 48-hour period. The errors shall not exceed those specified in 4.6.16. the indicator shall then be maintained at room temperature for a period of not less than 4 hours after which it shall be subjected to the scale error test specified herein. The errors observed shall not exceed the tolerance specified. The indicator shall be examined for deterioration or damage due to the extreme high temperature exposure.

4.6.23 Salt fog test. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810C, Method 509.1. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerance after the salt fog test shall not exceed the room temperature tolerances specified in table III. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including electrical receptacle, shall be cause for rejection.

4.6.24 Thermal shock test. The indicator shall be subjected to a total of eight cycles of immersion in tap water at a temperature of 85° ±2°C and then in tap water at a temperature of 5° ±2°C. The length of time for each bath immersison shall be 30 minutes with not more than 3 seconds between immersions. The indicator shall then pass the test specified in 4.6.7.

4.6.25 Shock test. The indicator shall be subjected to a shock test in accordance with MIL-STD-810C, Method 516.2, procedure I, except the shock force shall be 15g. Immediately after this test the indicator shall be subjected to the room temperature scale error test. The error observed shall not exceed the tolerance specified for the applicable indicator.

4.6.26 Electromagnetic interference test. The electromagnetic interference test shall be conducted in accordance with MIL-STD-462, except that the frequency range shall be from 0.15 to 150 mc. Test methods shall be those required in MIL-STD-461 for class A1, category b equipment.

4.6.27 Vibration endurance test. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810C, Method 514.2, equipment category (b), curve B,

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except from 5 to 12.5 Hz, the vibration shall be 0.250 inch double amplitude and shall be 2 g from 12.5 to 500 Hz. The indicator shall be operating and shall be vibrated in each of the three mutually perpendicular planes with a signal input off approximately half scale. Following the vibration period, the indicator shall be subjected to the scale error, friction error, and hysteresis tests. Room temperature tolerances shall be within those specified, and no screw or other parts shall become loosened or damaged as a result of the applied voltage.

4.6.28 Lighting tests. The indicator shall be subjected to all the tests specified in MIL-L-25467 for red-lighted indicators and to all the tests specified in MIL-L-27160 for white-lighted indicators.

4.6.29 Compatibility tests. Compatibility tests shall be conducted as required to demonstrate compliance with 3.5.1, 3.5.2, 3.5.3, 3.5.10, 3.5.11, 3.6e, 3.6.4, and 3.6.5. The EFU-26/A series indicators shall be properly connected to their respective transmitters which have been filled with fuel and capped off. The lowest specified voltage and frequency shall be applied for 100 on-off cycles. The transmitter motor shall not fail to start during any cycle.

4.6.30 Reliability test. The seven indicators specified in 4.5.2.4 shall be subjected to a reliability test in accordance with MIL-STD-781C, test plan IC.

4.6.30.1 Test equipment. The test equipment and fixtures shall include provisions for indicator operation cycling, temperature cycling, and vibration.

4.6.30.2 Operation cycling. During this test, the indicator shall be continuously cycled from 5 to 90 percent of full scale and return at a rate of one cycle every 4 minutes.

4.6.30.3 Temperature cycling. Unless otherwise specified or amended by the results of the thermal analysis, the temperature of the chamber in which the indicators are tested shall be cycled as follows:

a. Starting at room temperature, decrease temperature to -54°C and stabilize for 1.5 hours with indicators unenergized.

b. Turn indicators on and raise temperature to 55°C and operate indicators for 4 hours.

c. Turn indicator off and lower chamber temperature to -54°C for 2 hours.

d. Repeat steps b and c until reliability requirement has been met. This schedule provides four 6-hour test cycles for each 24-hour period. In each 24-hour period, the operational test time is 16 hours.

4.6.30.4 Vibration cycling. Throughout this test, the indicators shall be vibrated for a period of 10 minutes during each hour of on-time at any nonresonant frequency between 20 and 60 Hz and at a peak acceleration of 2.2 g ±10 percent.

4.6.30.5 Performance check. At each 200-hour interval and at the end of this test, the indicators shall be subjected to the following performance tests: Scale error, friction error, hysteresis, response time, damping, and lighting (color and brightness tests). During each scale error test, if any two test points exceed the allowable temperature-endurance tolerances specified in table I or III, as applicable, one failure shall be charged to this test. The lighting circuit shall be energized during this entire test. The indicators shall meet the lighting

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tolerances of the applicable lighting specification for the first 1,000 hours of testing on each indicator.

4.6.30.6 Failure criteria. For an accept decision, each indicator shall be operated at least 1,000 hours. If an indicator fails, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the indicators still under test. The contracting activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781C, test plan 1C, after each indicator has been operated for at least 1,000 hours.

4.6.30.7 Reporting. Reports shall be submitted in accordance with MIL-STD-781. The reliability test log and the failure summary and analysis shall be included in the final test report (see 4.4.2).

4.6.31 Longevity test. Three of the seven reliability test samples shall continue to be tested through a longevity test until 2,500 operating hours have accumulated on each of the three indicators.

4.6.31.1 Evaluation of longevity test. No accept/reject criteria for this test is specified. The evaluation shall be limited to the analysis of all failures, identification of design and pattern failures, and the submission of recommended design changes to the contracting activity.

4.6.32 Maintainability test. The contractor shall submit a maintainability plan to the contracting activity for review and approval to determine that the requirements specified in 3.5.14 have been met.

4.6.33 System safety. Identification of system safety hazards shall be accomplished by conducting a subsystem hazard analysis according to the fault hazard analysis specified in AFSC DH1-6, Chapter 2, safety analysis procedures and MIL-STD-882.

4.7 Inspection of preparation for delivery. Test methods for preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification. Packing shall be accomplished as specified herein.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity,

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each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-I of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness of insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicators shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicators packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicators shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The EFU-26/A and EFU-43/A indicators are intended for use in displaying rate of fuel flow in pounds per hour being consumed by the aircraft engine. The EFU-26/A indicators also supply power to the transmitter motor.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Type of indicator required (see 1.2).

c. That the preproduction test procedures are to be submitted to the contracting activity for formal approval before the test program is initiated (4.4).

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d. That the contractor shall submit a test box of wiring harness to the contracting activity for laboratory testing.

e. When the preproduction test samples are not to be retained by the contracting activity (see 4.4.1.1).

f. When the tests are not to be conducted in the order listed under 4.6.

g. When sampling plan B tests are not to be conducted (see 4.5.2.2.).

h. That the items specified in 4.5.3 be returned to the contractor for appropriate corrective action at the contractor's expense.

i. Levels of preservation and packaging, and packing required (see 5.1 and 5.2).

j. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL). The data produced by this specification are as follows:

a. A preliminary list of electronic parts as specified in 3.3.

b. An outline drawing of the indicator showing dimensions, displays, et cetera, and all other configuration details required to demonstrate complaince with this specification.

c. An internal circuit schematic and interconnecting diagram.

d. An installation instruction sheet.

e. Three copies of the test report specified in 4.4.2, signed by the Government inspector, demonstrating complaince of the indicator with all tests specified herein.

f. When required, test data as specified in 4.5.2.2.1.

g. Detailed reliability test program as specified in 4.5.2.5.1.

h. Procedures for conducting the compatibility tests are required by 4.6.29.

i. Detailed analysis of a failure as required by 4.6.30.6.

j. Reports as required by 4.6.32.

k. Maintainability plan (see 4.6.32).

l. Subsystem hazard analysis (see 4.6.33).

6.4 Definition. For the purpose of this specification, the following definition will apply:

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6.4.1 Hermetic seal. A hermetic seal is a perfectly closed, airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets or O rings.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Govenment tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/MMEDO, Tinker Air Force Base, OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.6 Subject term (key word) listing.

Fuel
Indicator
Rate of Flow
Type EFU-26
Type EFU-43
0-15000 PPH

6.7 International standardization. Certain provisions of this specification are the subject of international standardization agreement, STANAG 3872 and ASCC 10/69. When amendment revision or cancellation of this specification is proposed which will modify the international agreements concerned the preparing activity will take appropriate action through international standarization channels including departmental standardization offices to change the agreement or make other appropriate accomodations.

Custodians:

AIR FORCE -99
NAVY -AS

Preparing Activity:

AIR FORCE -71

Reviewer:
DepSo -01

Project No:
6620-0471

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